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RULix: A Foundation Model for Zero-Shot Remaining Useful Life Prediction

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Abstract

Maintaining manufacturing machines is costly and hard to schedule: operators must reserve a maintenance window, procure replacement parts, and decide which machine to service first. Data-driven remaining-useful-life (RUL) estimation predicts when a machine will fail and enables planning ahead, yet it requires run-to-failure histories that production assets rarely provide. Preventive maintenance truncates degradation before failure, and fleets are small, so labeled degradation data remain scarce at any single site. Recent studies therefore repurpose time-series foundation models for prognostics, while pretraining on degradation processes remains largely unexplored. This paper presents RULix, a prognostic foundation model pretrained exclusively on synthetic run-to-failure data. A synthetic prior generates sensor data from degradation processes based on stochastic differential equations (SDEs) and observed through structural causal graphs unrolled over time. An extended long short-term memory (xLSTM) encoder embeds sliding sensor windows in a latent space that represents machine dynamics, from which we predict the machine's health state, degradation dynamics, and RUL. The pretrained model is evaluated zero-shot on the C-MAPSS and N-CMAPSS turbofan benchmarks. The approach is a step toward prognostic foundation models for assets without recorded failure histories.

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Keywords: Predictive maintenance; Remaining useful life; Foundation models; Synthetic data; xLSTM

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¹ These authors contributed equally to this work.

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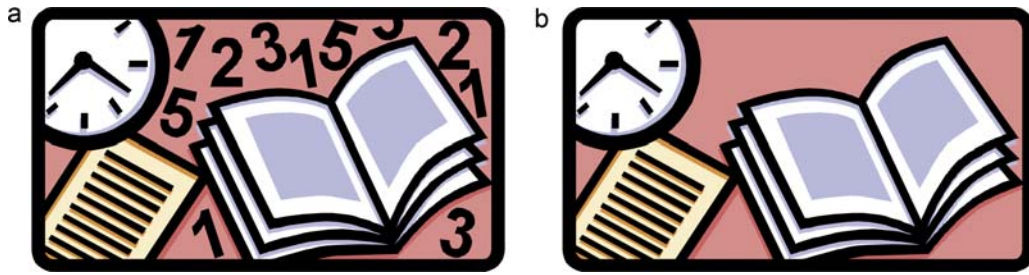


Fig. 1. (a) first picture; (b) second picture.

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$$\rho = \frac{\vec{E}}{J_c(T = \text{const.}) \cdot \left(P \cdot \left(\frac{\vec{E}}{E_c} \right)^m + (1 - P) \right)} \quad (1)$$

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References

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